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Department of Cyber Security

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CS23533 – FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

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CRIME INVESTIGATION EXPERT SYSTEM

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Introduction

Crime investigation requires analysis of crime records, suspect information, evidence and witness statements. The project develops an AI-based expert system to assist investigators using structured data and classical AI techniques.

Domain: Artificial Intelligence / Crime Investigation / Decision Support

- Reduces manual effort in investigation
- Organizes investigation data systematically
- Supports suspect analysis using AI techniques
- Provides transparent and explainable decision support

Problem Statement

Traditional crime investigation relies heavily on manual analysis of evidence, suspect information and witness statements.

- Large amount of investigation information
- Time-consuming manual analysis
- Dependence on investigator experience
- Limited automated suspect elimination and reasoning

A structured system is needed to process investigation information consistently and reduce human effort.

The proposed expert system integrates data management, search, constraints and AI reasoning to support investigation decisions.

Motivation

- To apply Classical AI concepts to a real-world problem
- To explore explainable rule-based decision support
- To reduce manual investigation effort

Research in police-assistance and crime knowledge systems shows the value of knowledge representation and rule-based reasoning.

Faster information processing, systematic suspect analysis, explainable conclusions and easier report preparation.

The modular design can be extended with additional crime types, evidence and investigation rules.

Objectives

1. To develop an AI-based expert system for assisting in crime investigations.
2. To represent crime knowledge using Predicate Logic and First-Order Logic.
3. To search possible suspects using Search Algorithms (BFS & DFS).
4. To eliminate invalid suspects using Constraint Satisfaction and Backtracking.
5. To identify the prime suspect using Forward Chaining and explain the reasoning using Backward Chaining.
6. To generate a comprehensive investigation conclusion report with logical justification.

Literature Survey

S.No.	Author & Year	Method / Technique	Key Findings	Limitations
1	Fernández-Basso et al., 2024	Rule-Based AI, Knowledge Base, NLP	AI knowledge-based system for police assistance in crime investigation.	Depends on structured knowledge and investigation rules.
2	Mohd et al., 2017	Ontology, Knowledge Representation	Models crime entities and relationships using knowledge representation.	Knowledge modelling can become complex for broader crime domains.
3	Agarwal et al., 2019	NLP, Rule-Based Extraction, Knowledge Base	Builds a knowledge base for crime entities and relationships from online newspapers.	Extracted online information requires preprocessing and validation.

Existing System

- Investigations rely heavily on manual analysis by investigators.
- Crime records and evidence are reviewed separately.
- Decisions depend largely on investigator experience and expertise.

Advantages:

- Human expertise can handle complex contextual evidence.
- Existing crime records can be retained for reference.

Limitations:

- Time-consuming manual analysis
- High human effort and possible inconsistency
- Limited automated suspect ranking
- No integrated AI reasoning or explainable decision support



Existing workflow: evidence and records are analysed manually, then the investigator makes the final decision.

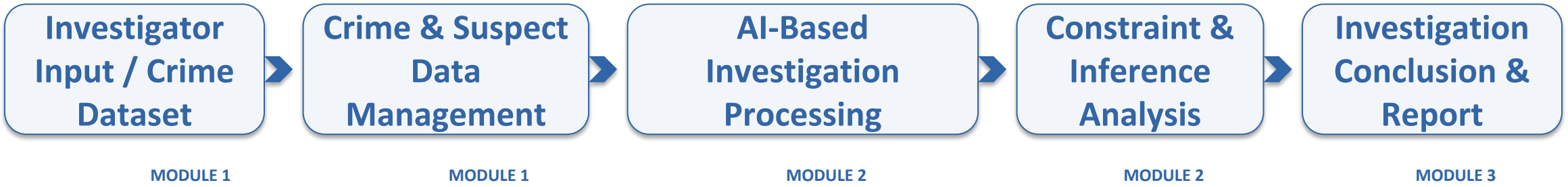
Proposed System

- Stores crime records, suspect profiles, evidence and witness statements in SQLite.
- Represents investigative knowledge using Predicate Logic and First-Order Logic.
- Uses BFS/DFS to search possible suspects.
- Uses CSP and Backtracking to eliminate invalid suspects.
- Uses Forward and Backward Chaining for inference and explanation.
- Generates an Investigation Conclusion Report.



Module 1: data management → Module 2: suspect search & elimination → Module 3: inference, explanation & final report

System Architecture / Block Diagram



Methodology

1. Data Collection / Investigator Input

2. Data Storage & Preparation

3. Knowledge Representation

4. BFS / DFS + CSP / Backtracking

5. Suspect Analysis & Validation

6. Investigation Conclusion / Report

Technologies Used

Programming Language: Python 3.11+
Framework / GUI: Tkinter
Database: SQLite
Tools / IDE: Visual Studio Code
Python Libraries: tkinter, sqlite3, json
Hardware: Intel Core i3 or above, 4 GB RAM minimum, 20 GB free storage

Module Description

Module	Description / Work	Key Techniques	Limitations / Remaining Work	Status
Module 1	Collect, organize and store crime, suspect and evidence information. Create and manage the SQLite database.	SQLite, Python, Tkinter, structured data management	Requires complete and accurate input data; future integration with larger datasets may be needed.	✓ **COMPLETED**
Module 2	Search possible suspects and eliminate invalid suspects using investigation constraints.	BFS, DFS, Constraint Satisfaction, Backtracking	Effectiveness depends on correctly defined investigation constraints and relationships.	✓ **COMPLETED**
Module 3	Identify the prime suspect, explain the reasoning and generate the final investigation conclusion report.	Predicate Logic, First-Order Logic, Forward Chaining, Backward Chaining	Not yet implemented at Review 1; final inference and report generation remain.	☐ **TO BE IMPLEMENTED**

Preliminary / Expected Results

PRELIMINARY PROGRESS



Completed outputs:

✓ SQLite database structure
✓ Crime / suspect data management
✓ Suspect search and constraint-processing workflow

Final inference and report generation will be completed in Module 3.

Project Schedule

Activity	Module 1	Module 2	Module 3	Final Testing
Literature Survey	✓	✓		
Problem Identification	✓	✓		
Requirement Analysis	✓	✓		
System Design	✓	✓		
Module Implementation	✓	✓	☐	

Current Progress

- ✓ Literature survey completed
- ✓ Problem identification completed
- ✓ Requirement analysis completed
- ✓ System architecture designed
- ✓ Technology selection completed
- ✓ Module 1 – Crime Data & Database Management completed
- ✓ Module 2 – Suspect Analysis & Constraint-Based Investigation completed
- ☐ Module 3 – AI Inference, Explanation & Report Generation
- ☐ Complete system testing
- ☐ Final evaluation

Future Work

Complete Module 3 implementation

Conduct complete system experiments

Compare manual / conventional workflow with proposed AI-assisted workflow

Evaluate suspect analysis and system performance

Prepare final system, report and documentation

Conclusion

The Crime Investigation Expert System applies Classical AI techniques to assist in analysing crime and suspect information.

Integration of multiple Classical AI techniques in one modular and explainable investigation system.

Systematic suspect analysis, explainable reasoning and automated investigation conclusion generation.

References

C. Fernández-Basso et al., “An AI Knowledge-Based System for Police Assistance in Crime Investigation,” Expert Systems, 2024.

F. Mohd, S. M. Shamsuddin, and N. Mustapha, “Knowledge Representation Model for Crime Analysis,” Procedia Computer Science, 2017.

S. Agarwal et al., “Crime Base: Towards Building a Knowledge Base for Crime Entities and Their Relationships from Online Newspapers,” Information Processing & Management, 2019.